PEI Group 17

Daniel Simbe
Anderson Tembe
Giovanni Santos
Felizardo Samuel
Marcelo Junior
Keegan Azevedo

ApontaUA

Collaborative Academic Document Management System

Milestone 3 | Computer Engineering Project | 2026/2027

Supervision:

João Barraca
Catarina Silva

CONTEXT

Centralize

All DETI study materials in one place, organized by degree, course unit and topic.

Collaborate

Students and teachers publish, comment, rate and suggest improvements to materials.

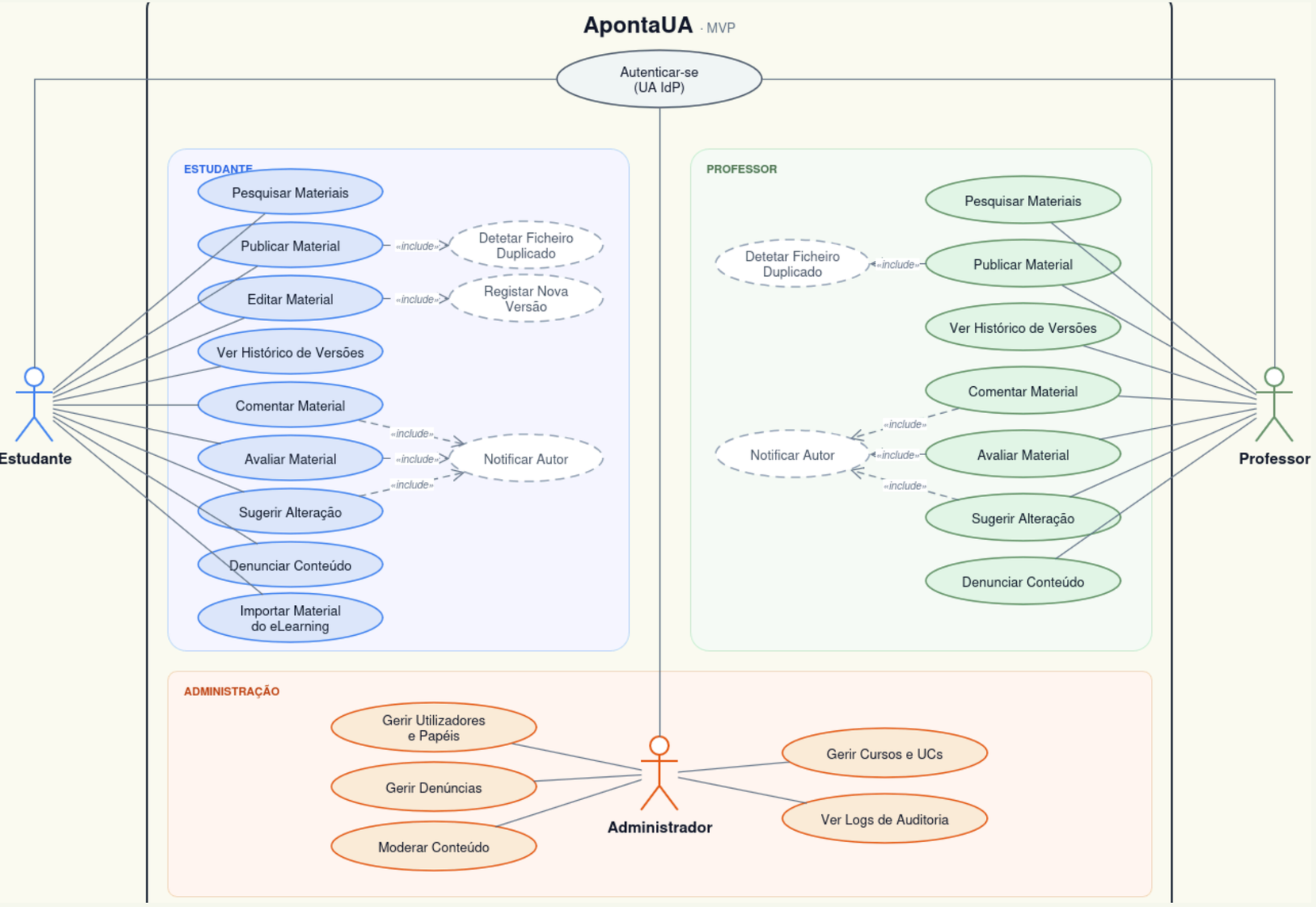
Ensure Quality

Version history, clear authorship, duplicate detection and content moderation.

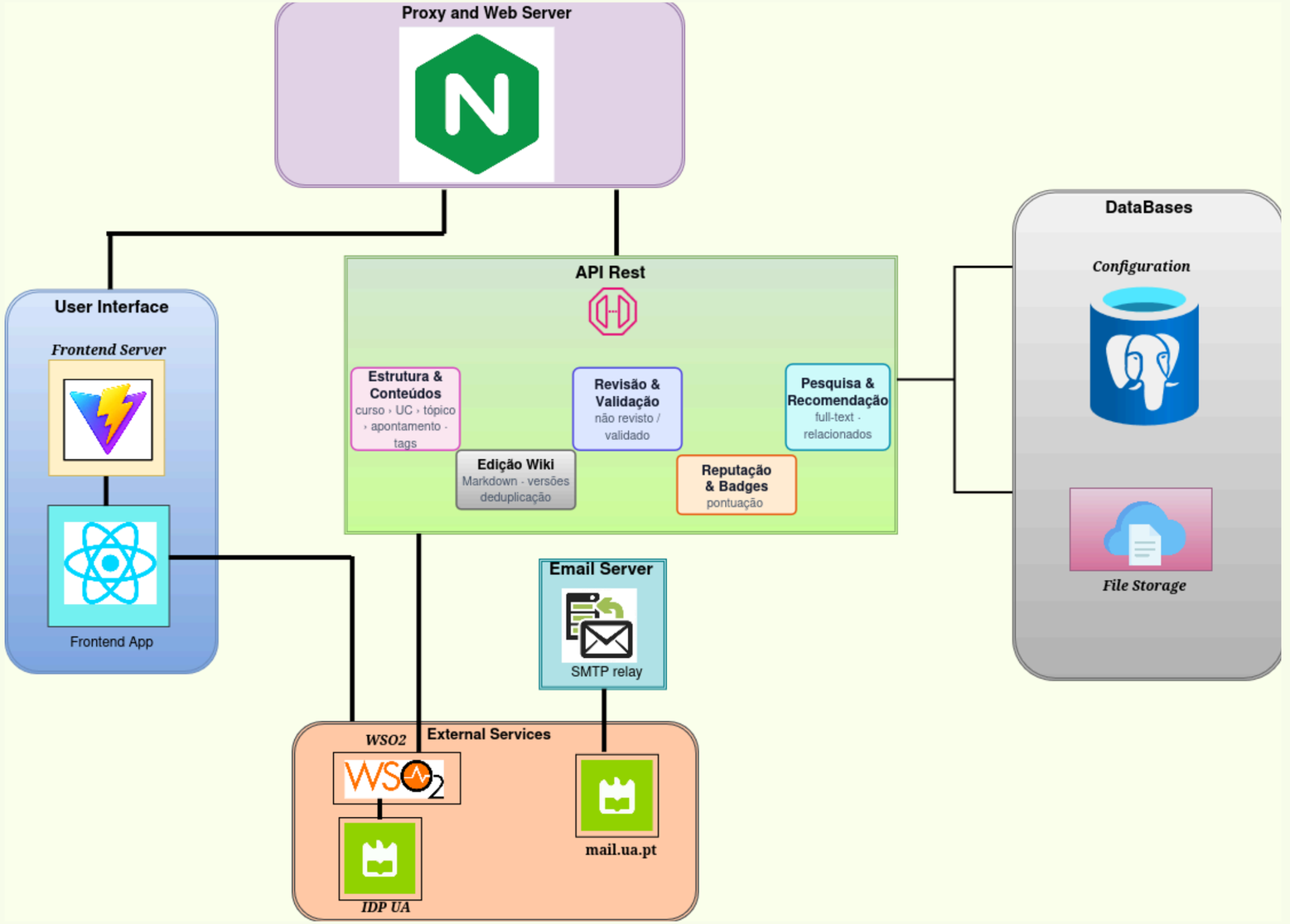
Reward Contributors

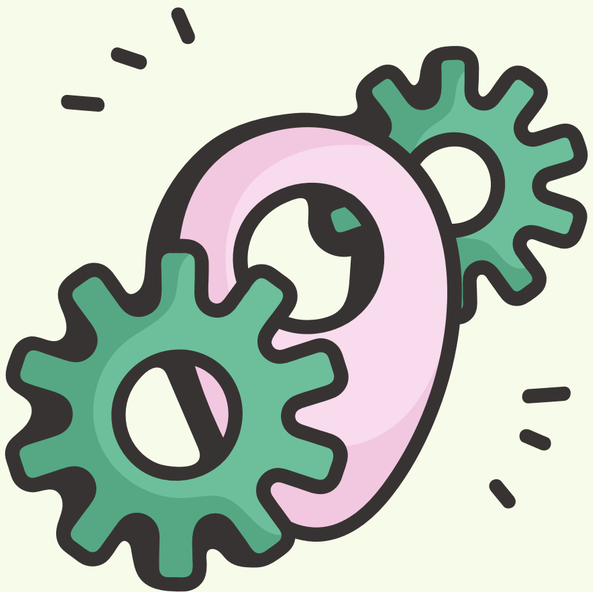
A reputation system that recognizes the most active and valuable contributors.

USE CASES


ApontaUA · MVP
Autenticar-se (UA IdP)
ESTUDANTE
Pesquisar Materiais
Publicar Material
Editar Material
Ver Histórico de Versões
Comentar Material
Avaliar Material
Sugerir Alteração
Denunciar Conteúdo
Importar Material do eLearning
«include»
Detetar Ficheiro Duplicado
Registar Nova Versão
Notificar Autor
Estudante
PROFESSOR
Pesquisar Materiais
Publicar Material
Ver Histórico de Versões
Comentar Material
Avaliar Material
Sugerir Alteração
Denunciar Conteúdo
Professor
ADMINISTRAÇÃO
Gerir Utilizadores e Papéis
Gerir Denúncias
Moderar Conteúdo
Administrador
Gerir Cursos e UCs
Ver Logs de Auditoria

SYSTEM ARCHITECTURE





TOPICS

1. RGPD
2. Licensing
3. Intellectual Property
4. Regulation
5. Terms and Conditions
6. External Dependencies
7. Technical Debt
8. Fragile Components and Risks

GDPR: Data

Data	Purpose	Retention
Internal identifier	Linking content	While the account exists
Institutional email	Notifications	While the account exists
Files and versions	Providing materials	Until removal
Queries and downloads	Measuring usage	12 months, then aggregated
Access logs	Detecting abuse	6 months

GDPR: Protection

PSEUDONYMISATION

Internal identifier

Visible pseudonym

Deleted user

SECURITY

HTTPS and encryption

Tokens validated in the backend

CNPD: 72 hours, where applicable

GDPR: DELETION AND THIRD PARTIES

Request > Verification > Deactivation > Review > Execution > Confirmation

Response: 1 month, as a rule

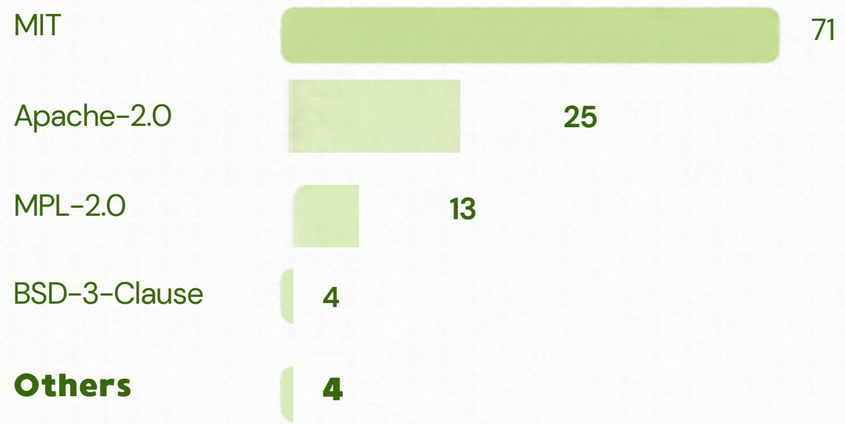
Backups: 30-day cycle

Third party	Data
Authenticated users	Name, materials, comments, reputation
UA (IdP)	Identifier and attributes
Hosting and email	Contract (Art. 28)

OPEN-SOURCE PACKAGES

120

packages (90 frontend + 30 backend)



No GPL, LGPL or AGPL

90 frontend, 30 backend

Pending: Docker images

INTELLECTUAL PROPERTY

Author keeps the rights

Licence limited to the service

eLearning # authorisation

Confirmation	Triage	Decision	Dispute
2	3	10	10

Internal targets, not legal ones.

REGULATION

APPLICABLE

UA Study Regulations

Copyright Code

GDPR and Law No. 58/2019

CONDITIONAL

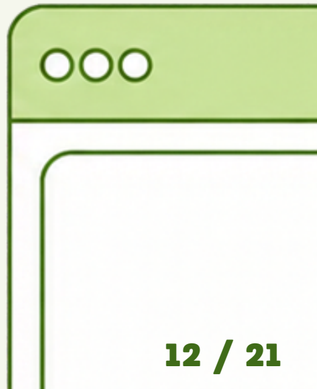
DSA: hosting service, platform status TBC

ISO/IEC 27001: security reference

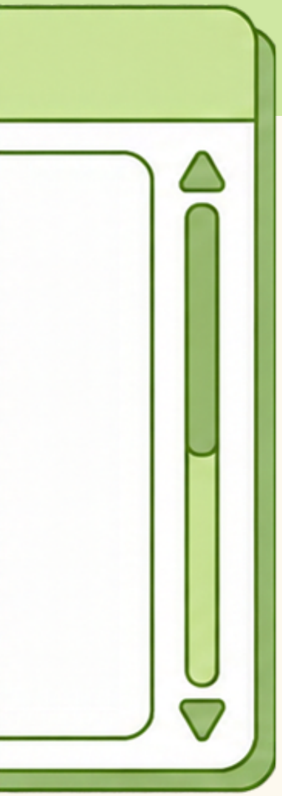
AI Act: not applicable (no AI used)

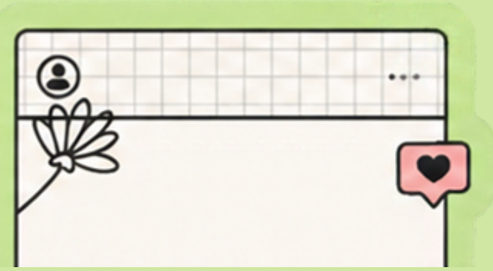
Terms and Conditions

1. **Use: lawful and academic**
2. **Content: users keep their rights**
3. **Disclaimer: no guarantee of accuracy**
4. **Moderation: with reasons and review**
5. **Changes: communicated to users**
6. **Law: Portuguese law applies**
7. **Acceptance: by use**



External Dependencies





Dependency	Risk	Mitigation
UA identity provider	No access granted	Isolated layer
PostgreSQL	Inconsistent data	Transactions and backups
Storage	Files without metadata	Orphan detection
Hosting at UA	Platform unavailable	Local copies

TECHNICAL DEBT & ACTION PLAN

Simulated authentication	Mock data and demo upload	No account deletion support	No automated tests
Integrate UA IdP	API, PostgreSQL and real upload	Anonymize content on deletion	Tests in CI

Fragile Components

Component	Impact	Mitigation
Authentication	No access	Isolated integration
Authorisation	Unauthorised access	Backend verification
Upload	Malicious content	Validate files
Consistency	Incomplete publications	States and recovery
Search	Materials hard to find	Define scope
Backups	Loss of materials	Test restore

RISKS AND MITIGATION

Risk = Probability × Impact

2 Critical

6 High

4 Moderate

Risk	Level	Mitigation
R1 IdP authorisation	12 Critical	Confirm with STIC
R2 Data loss	12 Critical	Test restore
R4 Malicious files	9 High	Validate files
R5 Inconsistent files	9 High	States and recovery
R3 Unauthorised access	8 High	Backend authorisation

ACCESSIBILITY MATRIX (WCAG 2.2 AA)

Screen · element	Affected users	Issue	WCAG criterion	Severity	Proposed fix
Notifications “Mark all as read” button	Low vision	Button border barely visible (1.21:1, needs 3:1)	1.4.11 Non-text Contrast	Medium	Darken border to #758A77 (3.47:1)
Search · category filters	Keyboard users	13 Tab presses from the search box to the first result	2.4.1 Bypass Blocks	Medium	“Skip to results” link
Administration “Suspend” button	Low vision, cognitive	Looks like plain text: no border or icon	1.4.11 Non-text Contrast	Low	Add #758A77 border or an icon

Low-contrast borders

PROBLEM

ApontaUA Início Explorar Pesquisar Publicar Contribuições 3 IM

Notificações

Tens 3 notificações por ler.

Marcar todas como lidas

Mostrar só as não lidas

Gerir notificações por email

Por ler O teu material «Guia de RGPD para projetos de software» recebeu uma avaliação de 5 estrelas.
Avaliação 30/10, 21:12
Abrir material Marcar como lida

Por ler Rui Carvalho pediu a revisão de «Guia de RGPD para projetos de software».
Pedido de revisão 29/10, 10:05
Abrir material Marcar como lida

Por ler Beatriz Lopes sugeriu uma alteração a «Resolução do exame de 2024/25».
Sugestão 23/10, 18:40
Abrir material Marcar como lida

Tiago Fonseca respondeu ao teu comentário em «Exame de época normal 2024/25».
Comentário 21/10, 09:30
Abrir material

A tua sugestão «Corrigir exemplo de mocks» foi aceite por Beatriz Lopes.
Decisão 8/10, 16:02
Abrir material

SOLUTION

Home > Resources > Contrast Checker

Foreground
Hex Value # 758A77
Color Picker Alpha 1
Lightness

Background
Hex Value # F6F8F5
Color Picker
Lightness

Contrast Ratio
3.47:1

permalink

Normal Text
WCAG AA: Fail
WCAG AAA: Fail
The five boxing wizards jump quickly.

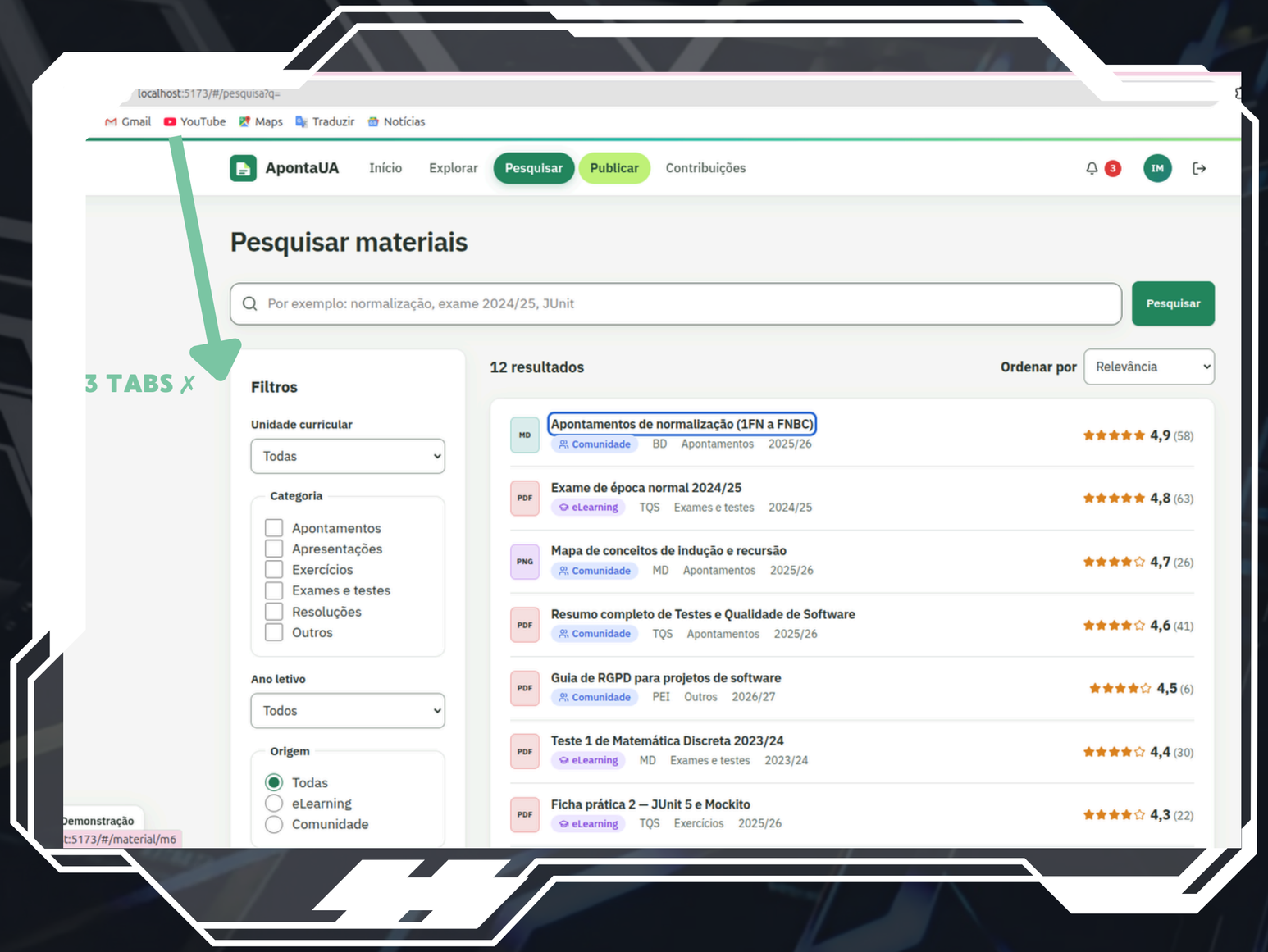
Large Text
WCAG AA: Pass
WCAG AAA: Fail
The five boxing wizards jump quickly.

Graphical Objects and User Interface Components
WCAG AA: Pass
Text Input

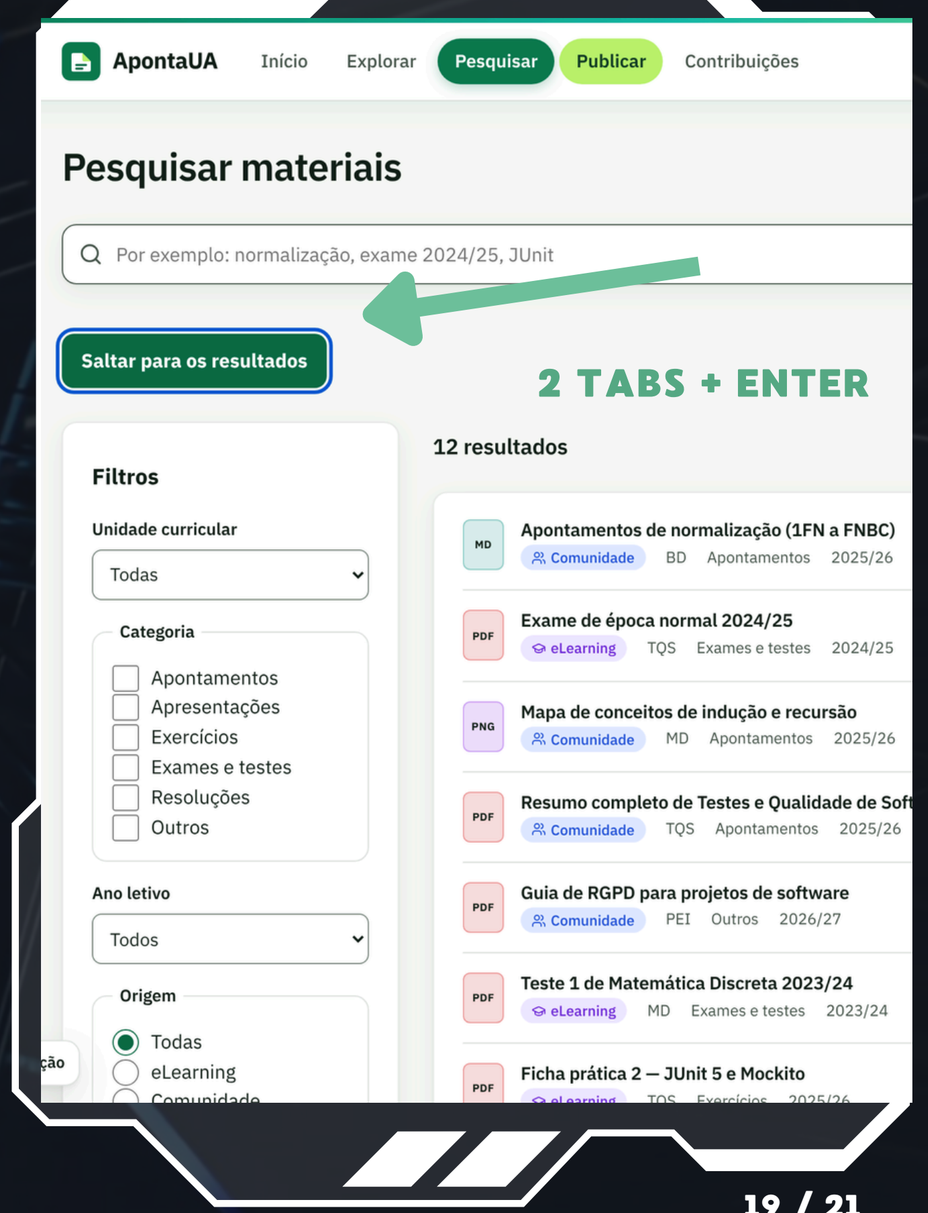
WebAIM Accessibility Testing Services

Tabs to reach the results

PROBLEM

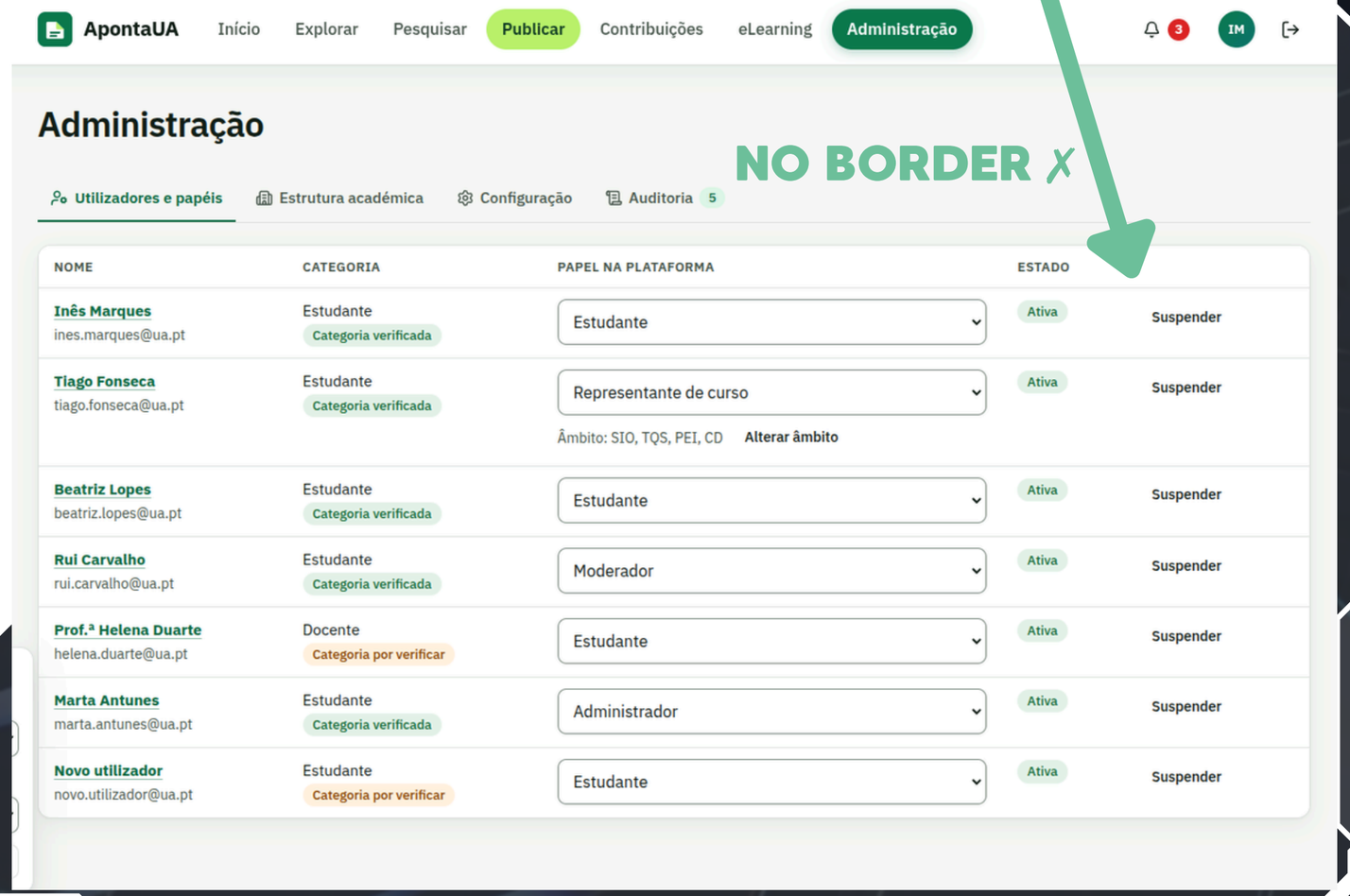


SOLUTION



Button without a visible border

PROBLEM



SOLUTION

Suspender

SAME FIX AS BORDERS
ADD A #758A77 BORDER
3.71:1 ✓

ApontaUA

THANK YOU FOR YOUR ATTENTION

https://apontaua.github.io/apontaUA-microsite/